MODERNISING DTU IDENTIFICATION SYSTEM

TechMeetUp

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INTRODUCTION OF PROPOSAL

In this era, where Artificial Intelligence, Industry 4.0 concepts, Machine Learning, Deep Learning, Mechatronics and various software technologies are being used to automate the workplaces as well as our homes, these automations help in better monitoring of the workplace and in performing difficult and hectic tasks in a much simpler and easy way. Using advanced technologies and making them amalgamating with the human resource will produce one of the most effective and efficient systems. So keeping in mind all the advancement going around the globe we have proposed a model on automation of one of the most prestigious university of our country '**DELHI TECHNOLOGICAL UNIVERSITY**'.

In this proposal, we planned

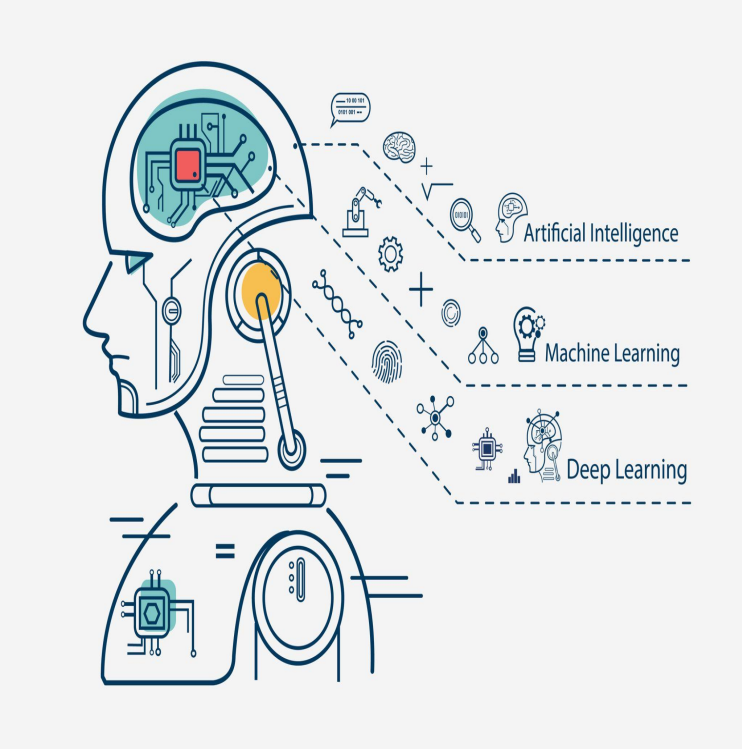
- On automation of entry and exit verifications of individuals who have an authentic ID card of university or are visitors to the university.
- To automate the license plate recording process of our university by using various machine learning tools and provide a pen and paper free entry-exit system at university's library and sports complex.
- On giving a better token system for keeping personal belongings at the shelves.

TECHNOLOGIES USED

The process of automation in the best way cannot be done by using a single technological aspect or software usage, it is a methodology in which wide aspects of technological and softwares uses are blended together to bring out a desirable output which will help in solving the challenges and taking up the task efficiently. The same is done with this idea of proposal.

In this project, we have used our existing knowledge on Machine Learning, Deep Learning, Hardware information and various other tools which would be effective in generating a result which is desirable.

So let's have a quick overview of the different technologies which we have used in this proposal for making DTU automate.



Machine Learning It is one of the most important tool in designing any automation process and using this gives an accurate result which helps in increasing the effectiveness of the technology. In this plan of DTU automate we have used various Machine Learning algorithms like Face Recognition System and Number Plate Detection.

Face Recognition System is used for detection of face of University Students with their photo which is being provided to the college to ensure that the one who is using the university identity card is the same to whom the Identity Card is being issued. It also helps us in keeping the record for any futuristic aspects.

Number Plate Detection is used to detect the license plate of any car entering The university and keeping the record of it with the individual driving it, if it is a Student's car, then it will be stored with the data of student entering the college.

BarCode Scanner is used which will read the barcode imprinted at the student's ID card to generate the information of verification and keeping the record that this identity card student has taken an entry or exit from the university.

Cameras are used everywhere to record all the things going around us and keeping the track of data which can be extracted easily at the time of need.

PROPOSED SOLUTIONS

- **RECOGNITE**
  It is a multi objective completion device. This device can scan the barcode, use a camera for face recognition algorithm, and also provide an interface for token system which we have introduced.

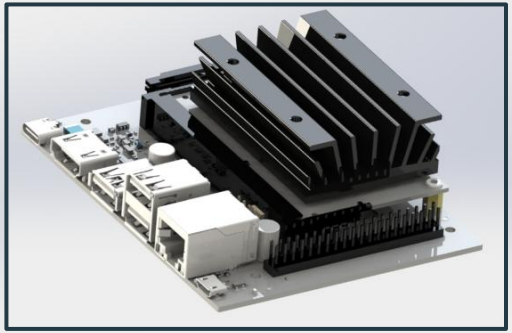
- **CARREAD**
  In this technology, we have used number plate detection algorithm with cameras to scan the number plate and keep their records.

RECOGNITE

Delhi Technological University
Security Check System
Please Scan College ID Card
2021-10-10
DELTECH

Components Used

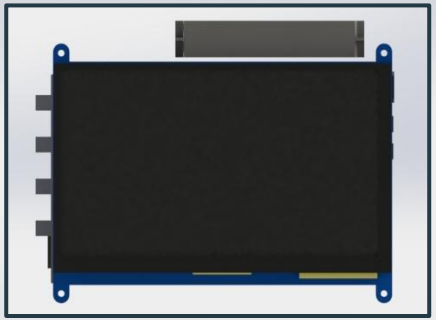
Cost: ₹5500
Weight: 250 gram

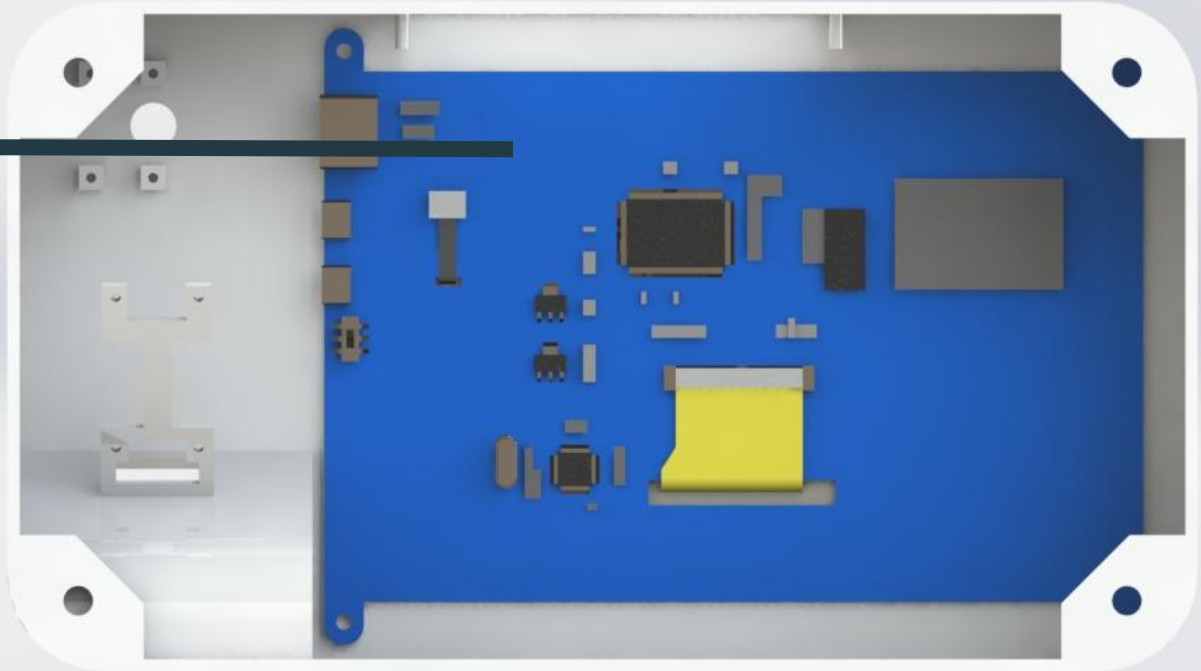


JETSON NANO

It is the Microcontroller used for all processing work. Since our system uses FRS for verification and NPD for car security system, NANO is capable of processing such large algorithms easily due to its 128-CUDA cores.

Cost: ₹2500
Weight: 500 grams

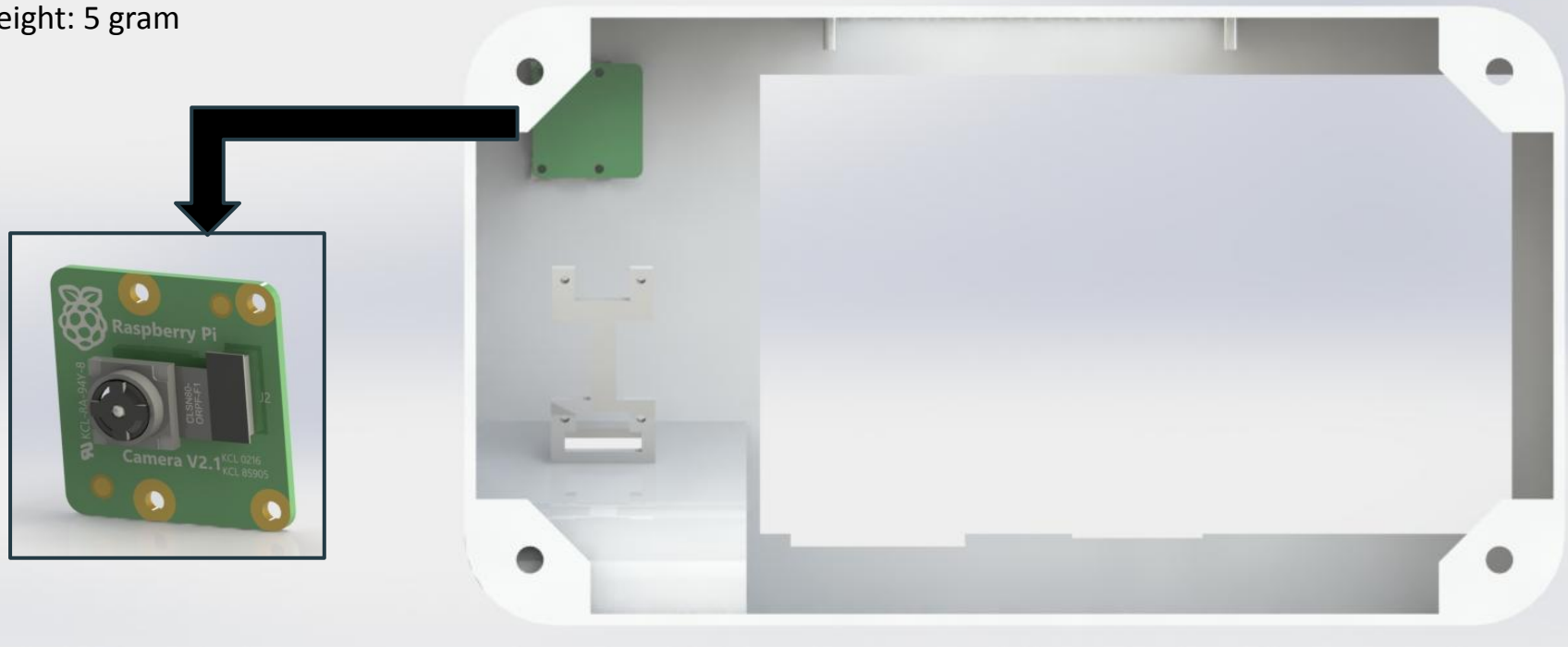




LCD SCREEN

This will act as an interface between the user and the system. It will show output after scanning the ID card and will show results for token number generated.

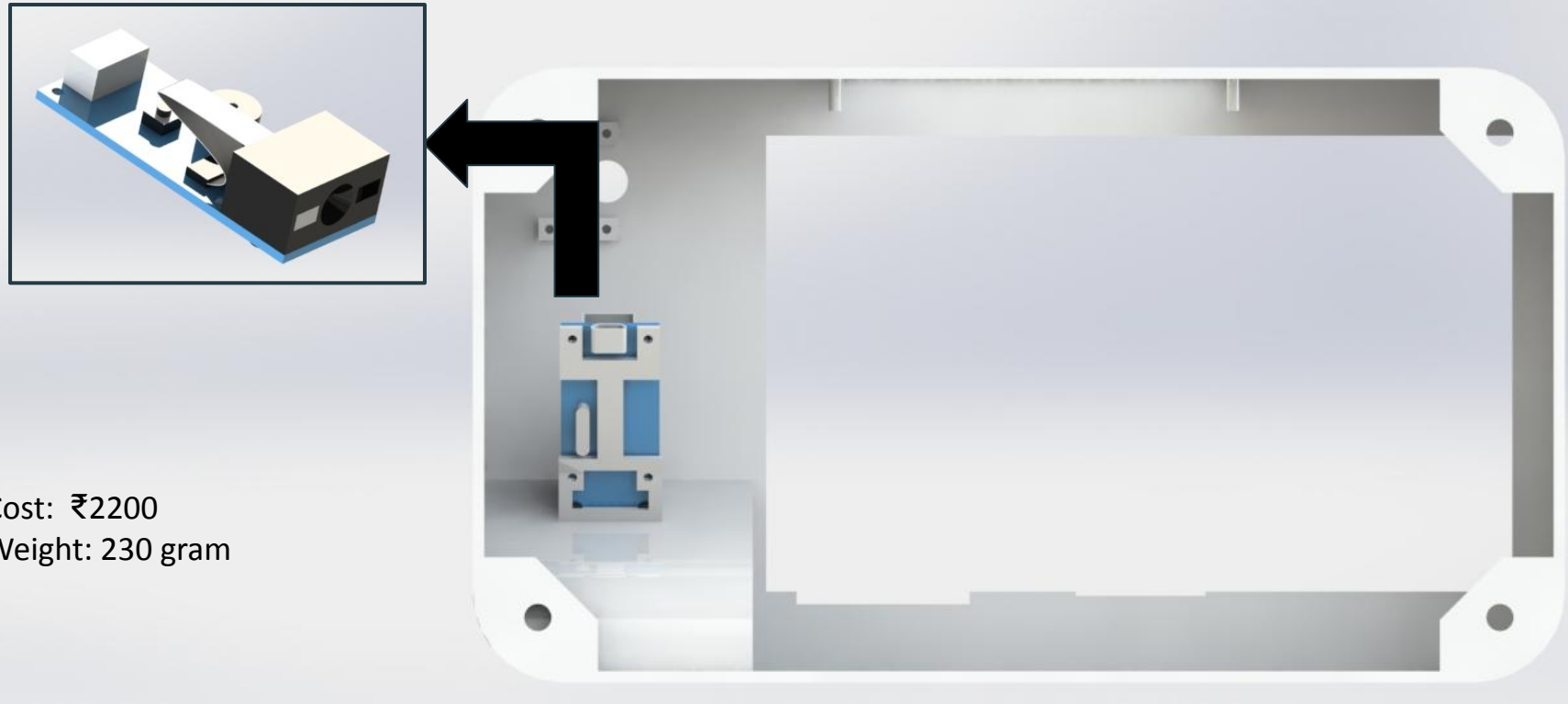
Cost: ₹1500
Weight: 5 gram



CAMERA MODULE

It is used for face recognition system. When identity card is scanned, then it will detect the face and using FACE RECOGNITION SYSTEM it will check whether the one using the ID card is the same person or not to whom the ID card is issued.

We have used Raspberry Pi Camera V2.1 which can work both in indoor and outdoor premises and is cost effective.

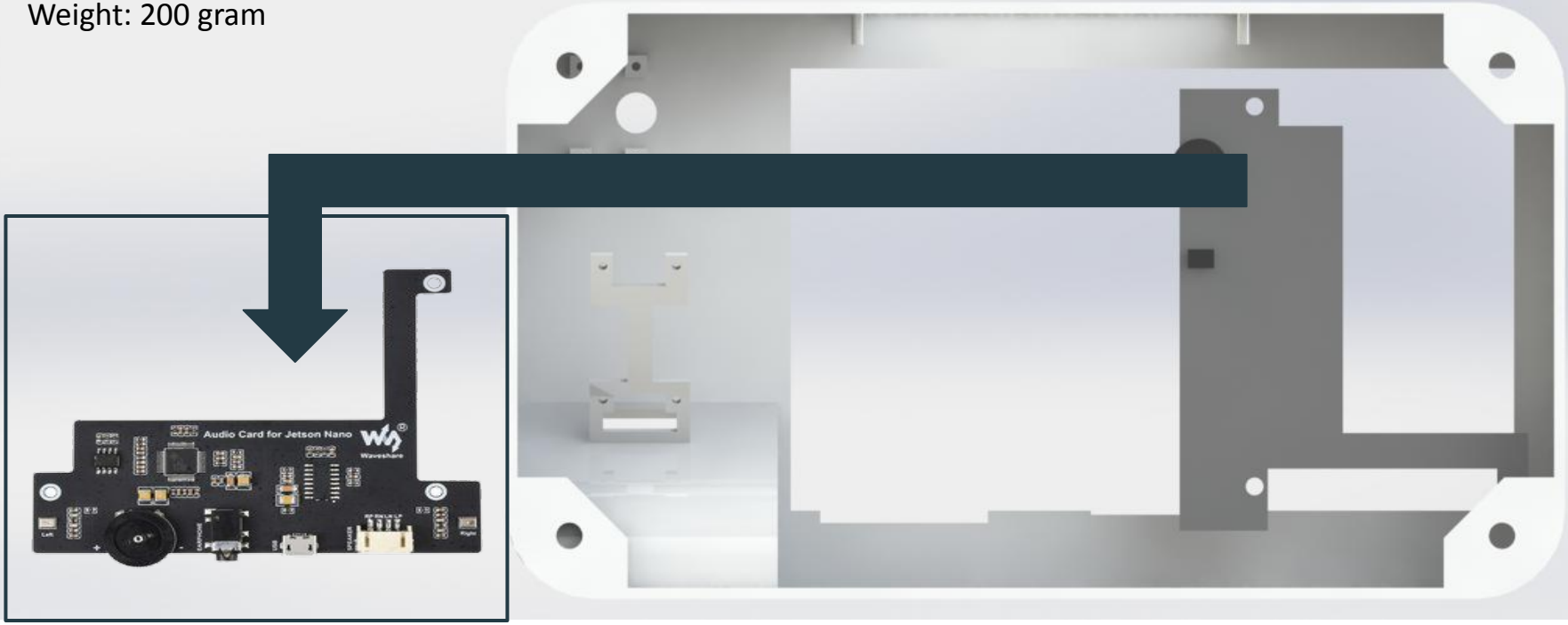


Cost: ₹2200
Weight: 230 gram

BARCODE READER

This sensor is used to detect the barcode printed on back of the ID card of the university and generate information and display on the LCD screen which are individual's name, ID card number and photo of the individual stored in the database.

Cost: ₹1000
Weight: 200 gram



AUDIO CARD

This is used as a microprocessor to generate sound which will help in understanding the individual whether their input is received or not. They are connected with speakers which produce sound.

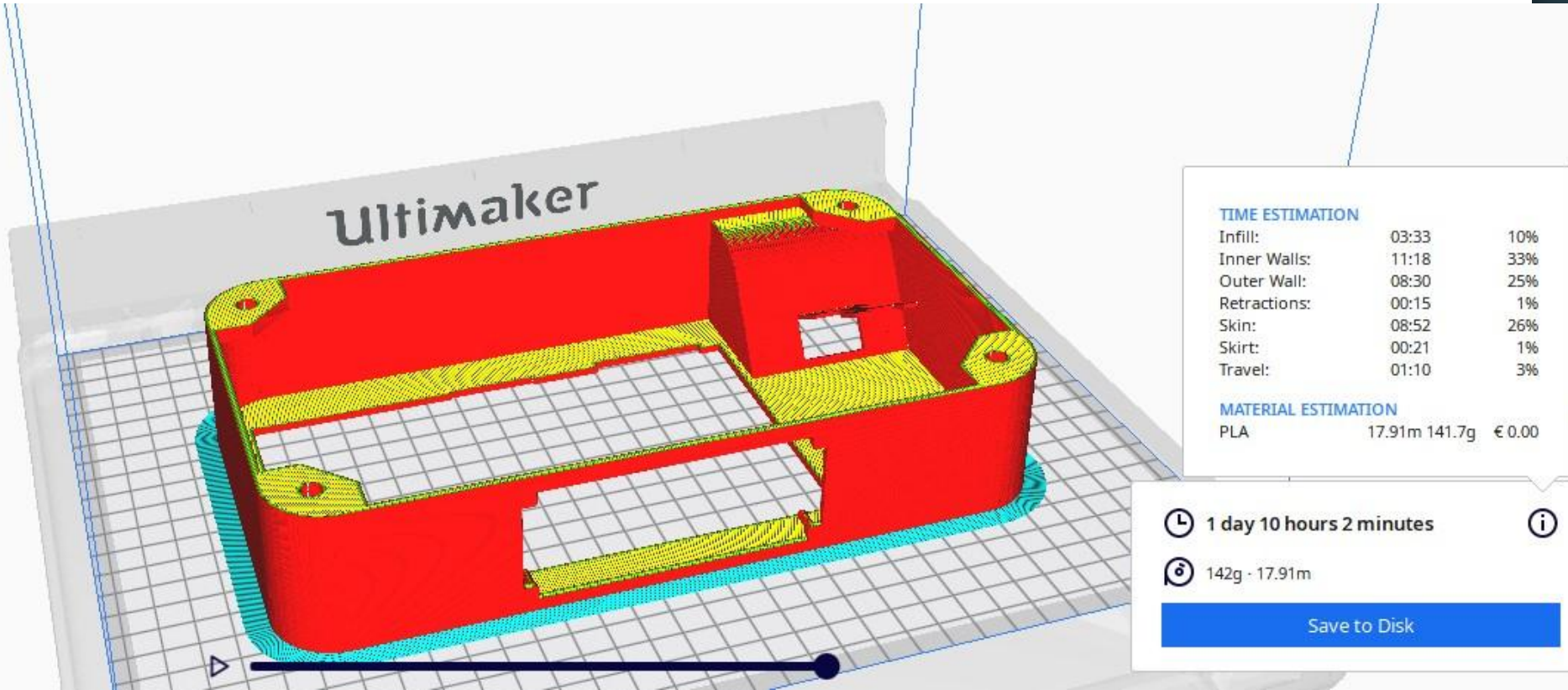
SPEAKER

They are connected with audio card and produce a sharp sound whenever an input is provided to the system and when system has generated some results.

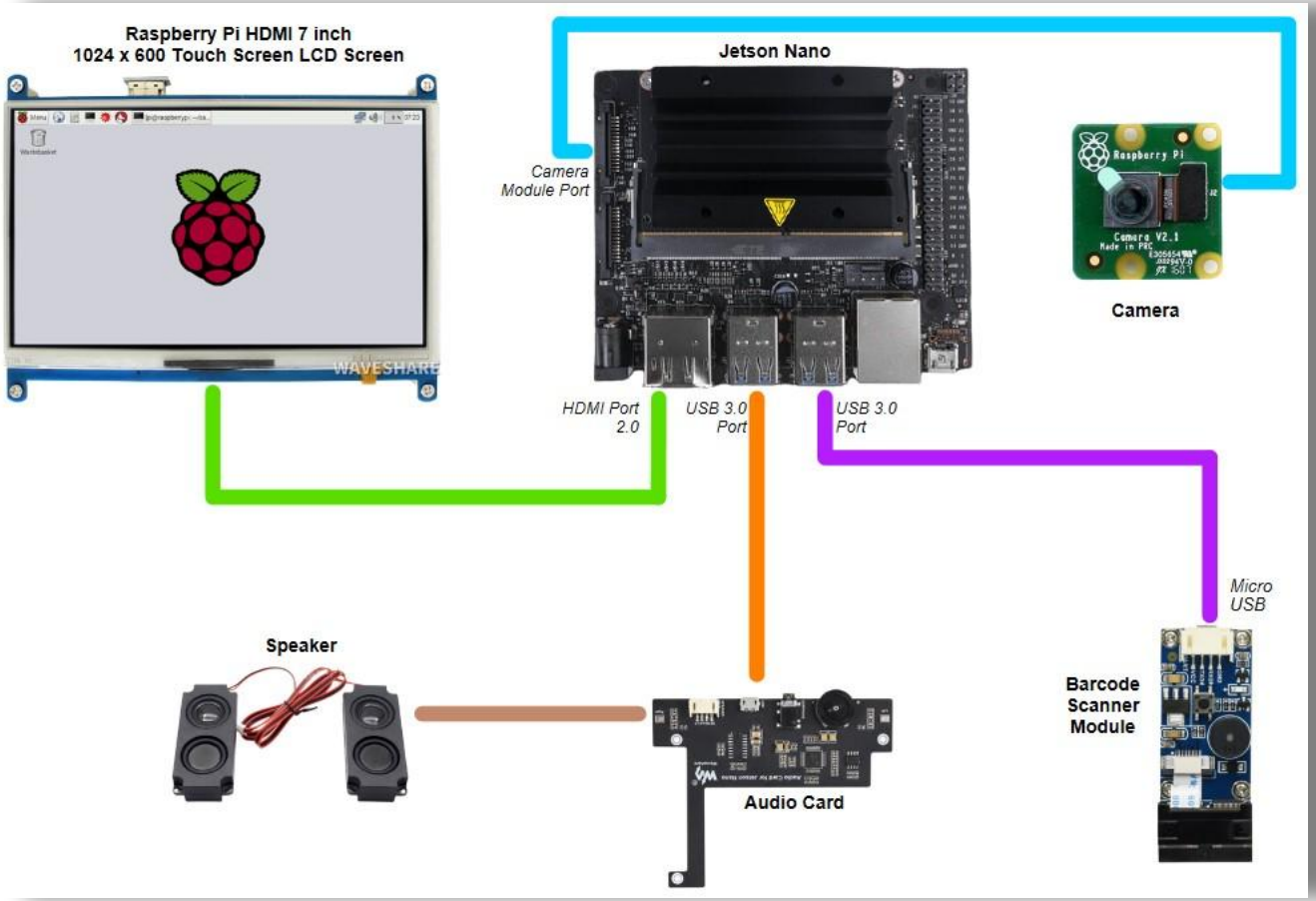
Total Material Used = 251.5 gram of PLA

Estimated Material cost = ₹300

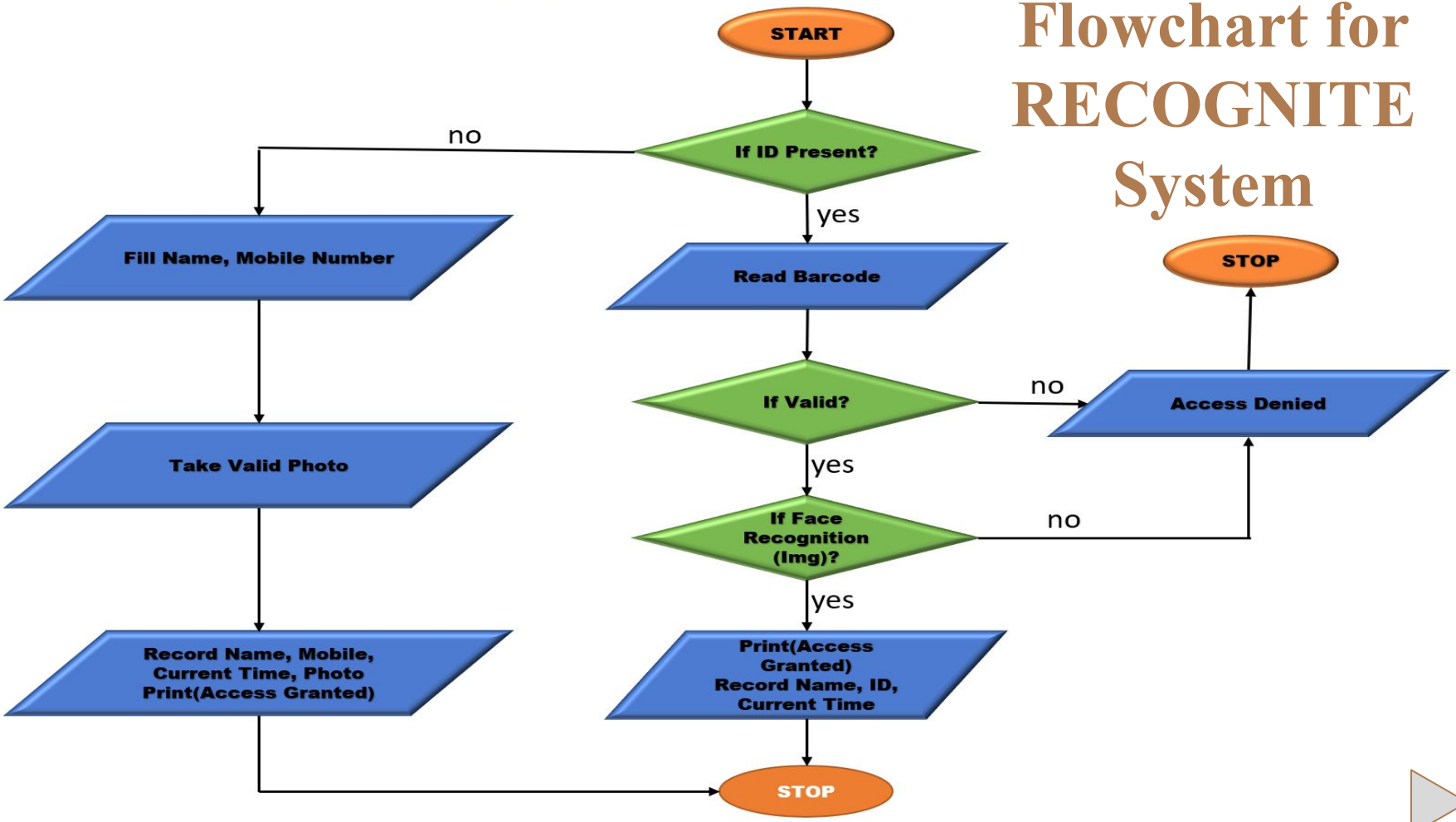
Estimated cost for 3D printing = ₹1500



CIRCUIT FOR RECOGNITE



Flowchart for RECOGNITE System



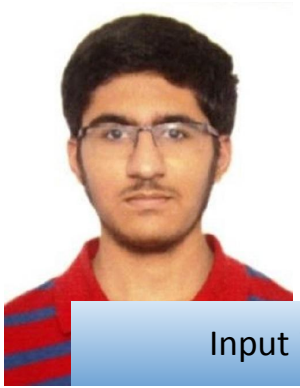
Face Recognition System

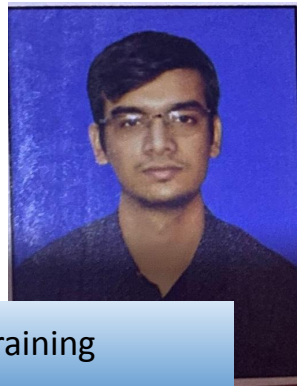
1. Creating **Known directory** for training images and **Unknown directory** for testing new unseen images.

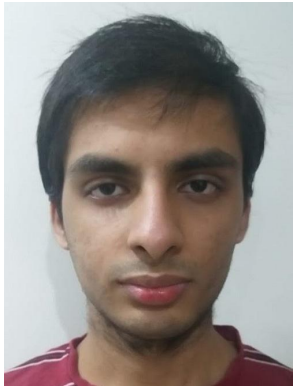
2. We used **OpenCV** to read images and the images are resized according to its initial ratio with width 500.

3. We used **image encodings** from ‘face_recognition’ library for each trained face and store them in a list named ‘known_encodings’ and the names associated with the files are stored in list named ‘known_names’.

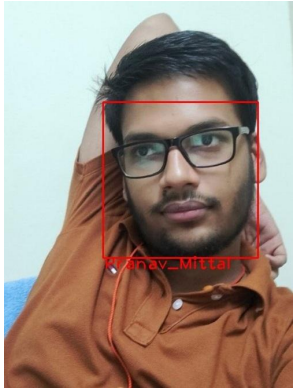
4. From Unknown directory, each file’s image encoding is extracted and is **compared** with the known encodings. If match is found, the face is detected using OpenCV and the name is printed below.

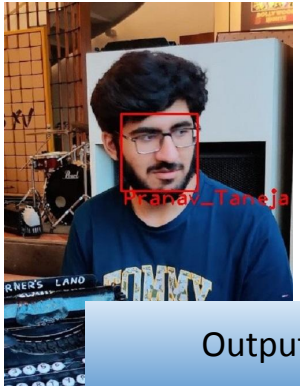






Input images for Training





Output images after Testing

COST OF RECOGNITE

A technological system which can solve so many challenges and with an optimum cost is a perfect balance.

The maximum cost of the system can be around ₹14,000 of one recognite.

The overall weight of RECOGNITE can be 1.6 - 1.8 kg

APPLICATIONS

There are a wide range of applications for this RECOGNITE system: -

1. It is used in normal days for verification of individuals who are entering or exiting from the university.
2. It can be used in events where huge crowd is gathered and can be used for verification purposes.
3. It can be used in entry and exit purposes in library and sports complex.
4. It can be used for generating tokens for keeping personal belongings in spaces provided by DTU.
5. It can be installed in DTU mess and DTU hostels.

CARREAD - Number Plate Detection System

COMPONENTS OF CARREAD

1- CAMERA- It is used for detection of number plate and scans it for the algorithm to work.

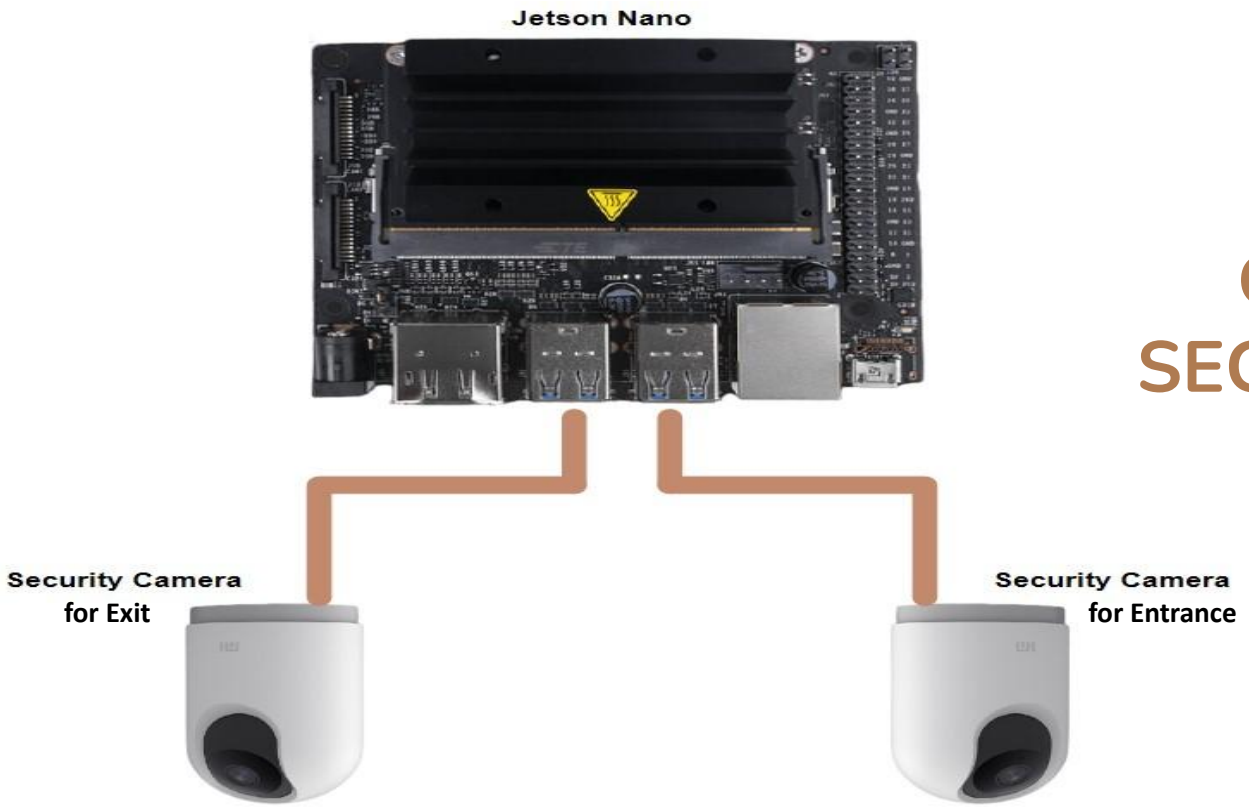
2- JETSON NANO- It is a microcontroller which will helps in multitasking and working efficiently.

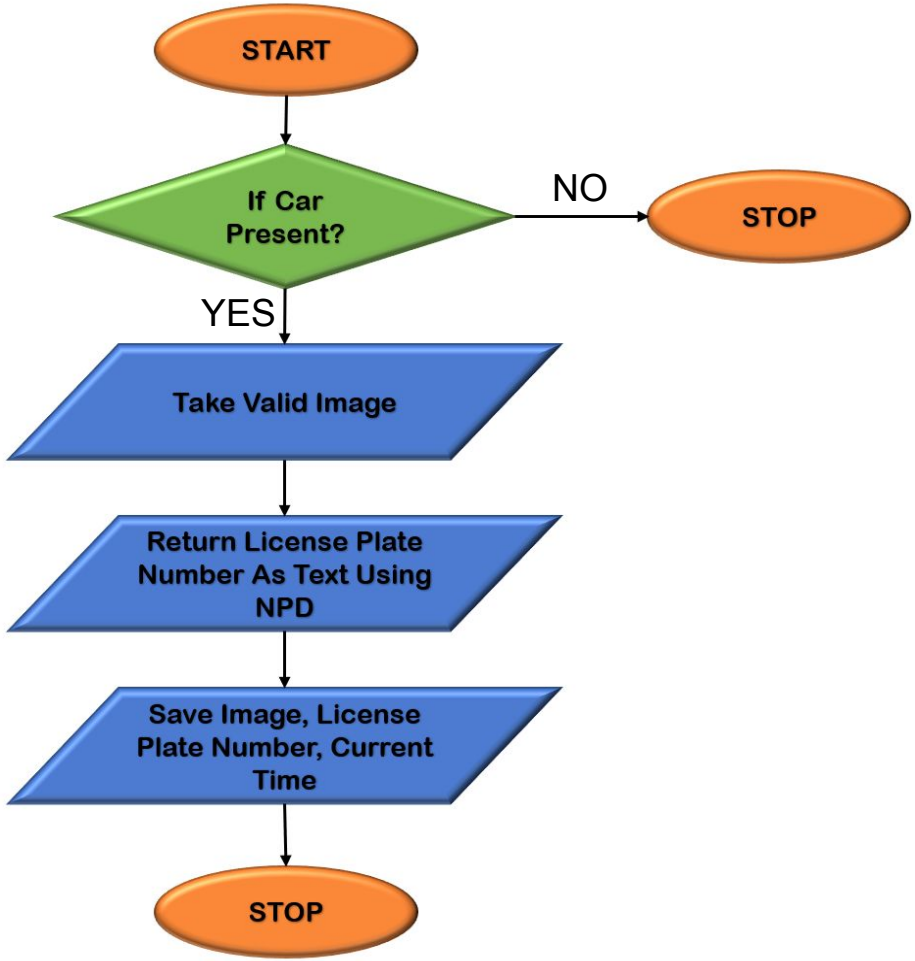
COST OF THE COMPONENTS

JETSON NANO - ₹5500

CAMERA - ₹2000

CIRCUIT FOR SECURITY SYSTEM





FlowChart for CARREAD System

Number Plate Detection Algorithm

1. We used **OpenCV** to read images, **cvtColor** function to make the image gray from colored and use matplotlib.pyplot to display the images.

2. We also used **bilateral filter** along with **Canny algorithm** to convert the gray scale image to an edged image.

3. We used OpenCV to find contours and image utility library (imutils) to make a contoured image and **ApproxPolyDP** algorithm helps us in finding the **location** of number plate in the image.

4. We extracted the area of number plate using **bit masking** and **numpy arrays** and we cropped that part out separately.

5. This cropped image is now fed to **readtext algorithm** from EasyOCR which returns the text output from the image.

Original Photo	Cropped License Plate	Text Output from EasyOCR
		MH 20EE 7598
		DL 3CBL 2856

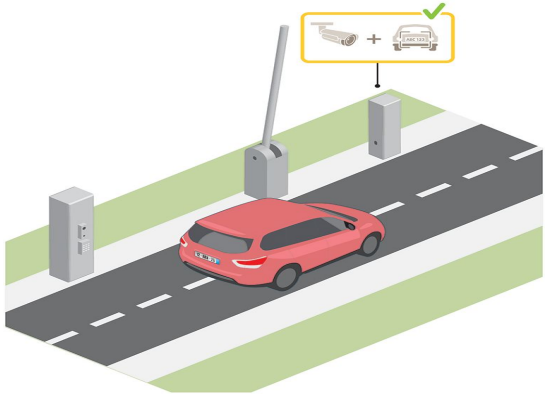
Applications

1- It is used to scan number plate of the cars taking entry in the university premises.

2- It also keeps the record of data of number plates scanned.

REVOLUTION AFTER IMPLICATION

- The old manual process in which guards used to look at every individual ID card will we replace.
- The accuracy for verification will increase and a proper data of individuals entering or exiting from the college will be kept and can be used when needed.
- Safe for environment as we are replacing the old methodology of pen and paper.
- Guards will now have more time for doing some other chores like increasing the frequency of patrolling inside the campus.
- This process will save time of each and everyone.



FEASIBILITY

Now we are going to discuss the feasibility of the project which is having a huge potential to solve the future challenges.

DURABILITY- The material we have used to design hardware is Polylactic Acid (PLA) it is a thermoplastic material. The requirement of service for apparatus is less frequent and material can last a long time.

PRODUCTION- This hardware can easily be produced by using 3D printing. We can also use other different methods to generate this and it is simple process. Mass production also does not generate any additional challenges.

WORTHINESS - A system which takes care of the most important aspects of security that is verification is getting revolutionized with the help of technology and making the process automated with not having a high cost is a must need system for any institution.

Why Our System?

- Such an integrated system is not available in the market using FRS, BARCODE SCANNER and INTERACTIVE INTERFACE.
- Mass production will reduce the price substantially.
- Additional Capabilities like FRS and NPD Algorithms.
- Easy assembly and disassembly.
- The codes can be altered by the manufacturers to redesign the purpose specifically.

CONCLUSION

This discussion gives us a detailed review of our new security system and how it can be implemented along with the hardware and software requirements, how the project is feasible and what are its applications and what will be the results after we implement this technology to automate our verification system. All these above mentioned topics have been discussed and shown by the help of simulations. We have designed a new system which can help in the process of verification of individuals and which has a wide range of applications.

FUTURE SCOPE

Technological development can be applied in field of security where we used manual methods for verifications but with the proposal of this idea we have tried to revolutionise the old methodology of verification with the new one where we are using Machine Learning algorithms and new Hardware design, this project itself is a complete package which can help in solving various challenges.

The work which can be done in future to make this idea more efficient and more effective can be maintaining a database of every individual entering and exiting form the premises and how many times he did it in a day. This idea can be used in car parking area where the car number which was scanned during the time of verification at the entry can be stored with the student's ID and can be used in times where identifying the one who brings the particular car in the premises.

We can also use it more efficiently when large crowd is gathered and how things can be planned to make the process more accurate and effective.

We can integrate authority actions which are restrict particular students from accessing the facilities of library and sports complex under certain circumstances.

We can also integrate it with the lock system.

Thus, this project has an immense scope and great potential which can help in dealing with the challenges of the future.

Thank you